

MARVIN DURANGO COGOLLO

Computational Modeling & Digital Twin Engineer

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PROFESSIONAL SUMMARY

Engineer with a Ph.D. in Engineering Technology, specializing in physics-based computational modeling and digital twins, with equal strength in simulation and hands-on experimentation. Builds first-principles and data-driven models — Multiphysics/CFD, dynamic system simulation, discrete-event simulation, and real-time state estimation — and validates them against experimental and operating data. Doctoral research developed a physics-based digital twin for fluid-power (hydraulic) systems; industry and applied work span a manufacturing-cell digital twin at Hitachi and discrete-manufacturing systems (robotic cells, conveyors, AGVs) in Purdue's Smart Learning Factory. Bilingual (English/Spanish); effective across technical, cross-functional, and instructional settings.

MODELING & TECHNICAL COMPETENCIES

Computational Modeling & Simulation: First-principles / physics-based modeling, dynamic system simulation, multiphysics & computational fluid dynamics (CFD), multiphase & particle-laden flow (CFD–DEM, Eulerian–Lagrangian), discrete-event simulation, reduced-order / lumped-parameter models

Digital Twins: Sensor-integrated real-time digital twins, simulation-driven decision-making, model–data synchronization, empirical validation against operating data, robot commissioning

Optimization & Machine Learning: Nonlinear optimization (gradient-based and steepest-descent methods, KKT conditions), design-space exploration; machine learning and deep learning (CNNs / transfer learning with VGG16), computer vision (YOLO, OpenCV)

Estimation & Control: Extended Kalman Filter (EKF) state estimation, Bayesian parameter regression, nonlinear system identification, sensor fusion, real-time control

Experimentation & Characterization: Test-bench design & instrumentation, data acquisition (DAQ) & sensor calibration, particle-size analysis by laser diffraction (Malvern Mastersizer), empirical model validation against measured data

Software & Tools: Ansys Fluent, Visual Components, Simulink, MATLAB, Unity 3D, Python, C#; SPH, FEM

Embedded & Data: Raspberry Pi, Jetson Nano, PLCs, IoT data pipelines, real-time monitoring dashboards; Serial / UDP / I2C

Domains: Smart manufacturing & Industry 4.0, fluid power (hydraulic & pneumatic), robotics & automation, multiphase systems

PROFESSIONAL EXPERIENCE

Hitachi America, Ltd.

R&D Intern — Industrial Innovation Laboratory (Smart & Digital Manufacturing)

Farmington Hills, MI

Jun 2023 – Jan 2024

- Developed a sensor-integrated digital twin prototype for a manufacturing cell, enabling simulation-driven decision-making and streamlined robot commissioning.
- Built discrete-event simulations of packaging processes and demand-forecasting scenarios to evaluate throughput and capacity under varying operating conditions.
- Programmed UR5 collaborative-robot tasks for hybrid manufacturing workflows, increasing capacity predictability by **89%** and reducing operator fatigue by over **50%**.
- Conducted market and patent analysis to identify opportunities in robotics and digital-twin technologies aligned to customer needs.

Purdue University — School of Engineering Technology

Graduate Researcher (Ph.D.)

West Lafayette, IN

Aug 2021 – Present

- Developing a physics-based digital twin for hydraulically actuated serial manipulators, coupling first-principles dynamic models with real-time sensor data for online state estimation and condition monitoring.
- Implemented a 4-state Extended Kalman Filter (position, velocity, chamber pressures) deployed on embedded hardware for real-time estimation; identified nonlinear (Stribeck) friction parameters from experimental data.
- Designed and instrumented a hydraulic test bench and validated model predictions against measured data (**< 7% tracking error**), demonstrating model–hardware fidelity across multiple operating conditions.
- Conducts multiphysics/CFD and SPH studies of fluid-power components to characterize internal flow behavior and inform reduced-order model selection.
- Contributes to Purdue's Smart Learning Factory, applying digital-twin and simulation methods to discrete manufacturing systems (robotic cells, conveyors, AGVs).

Purdue University — School of Engineering Technology

Instructor / Graduate Teaching Assistant

West Lafayette, IN

2021 – Present

- Instructor of record (Fall 2025) for Mixed Reality Smart Manufacturing and AI & Mixed Reality for Manufacturing; designed a project-based course in which students build digital twins of a full discrete-manufacturing line — robotic arms, conveyors, and AGVs — in Purdue Polytechnic’s flagship learning factory.
- Instructor / TA for Advanced Fluid Power (MET 334) and Pneumatic Control (MET 436), covering system modeling, instrumentation, and control.
- Developed instructional materials integrating simulation and XR for real-time visualization of system dynamics.

Earlier Research

- Graduate Researcher & Teaching Assistant, Universidad de los Andes (2017–2019): optimized hydrocyclone separators using coupled CFD–DEM (Eulerian flow with discrete-element particle tracking) for solid–liquid separation; measured particle-size distributions by laser diffraction (Malvern Mastersizer) to parameterize and validate the model.
- Visiting Scholar, Purdue University (2019–2020): designed and built a custom hydraulic test rig; CFD modeling of multiphase reservoir systems.

EDUCATION

Ph.D., Engineering Technology — Purdue University, West Lafayette, IN	2026
Dissertation: Physics-Based Digital Twin for Hydraulic Serial Manipulators · Advisor: Dr. José Garcia-Bravo	
M.S., Mechanical Engineering — Universidad de los Andes, Bogotá, Colombia	2019
Thesis: CFD–DEM optimization of hydrocyclones (multiphase solid–liquid separation)	
B.S., Mechanical Engineering — Universidad Tecnológica de Bolívar, Cartagena, Colombia	2015
Selected graduate coursework: Optimization (nonlinear / gradient-based methods, KKT conditions), Machine Learning & Deep Learning, Computational Fluid Dynamics, Finite Element Analysis.	

SELECTED PUBLICATIONS

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- Durango, M., & Garcia-Bravo, J. M. (2026). *Smoothed Particle Hydrodynamics for Internal Flow Simulation in Hydraulic Components: Capabilities, Limitations, and Open Challenges*. Journal of Hydrodynamics (accepted).
 - Durango, M., Garcia, Y., Duncan, S., & Garcia-Bravo, J. M. (2026). *A Physics-Informed Hybrid Digital Twin Framework with Mixed Reality Interface for Fluid Power Teleoperation: A Pneumatic Proof-of-Concept*. Engineering Science and Technology, an International Journal (under review).
 - Durango, M., & Garcia-Bravo, J. M. (2025). *Demonstration of a Digital Twin Framework for a Two-Actuator Hydraulic Application* (in press).
 - Durango, M., & Garcia-Bravo, J. M. (2024). *Physics-Based Digital Twin for Hydraulic Equipment*. DigiTwin Conference.
 - Durango, M., & Garcia-Bravo, J. M. (2023). *SPH as a Model for Hydraulic Components*. ASME Fluid Power Systems & Technology Division (FPMC).
 - Durango, M., Borders, E., Newell, B., Garcia, J., & Breidi, F. (2022). *Mixed Reality for Fluid Power Instruction*. ASEE Annual Conference.
 - Durango, M., Garcia-Bravo, J., Newell, B., & Gonzalez-Mancera, A. (2020). *CFD Modeling of Hydrocyclones*. Fluids, 5(3), 118.